



Material Safety Data Sheet

Revision Date 29-Oct-2013

Version 1

1. PRODUCT AND COMPANY IDENTIFICATION

Product Label Name Kirkland Signature Ultra Clean Single Dose Laundry Pacs - 120ct
Product name KIRKLAND SIGNATURE 1/120CT SDL RFCPSC
UPC Code 10096619067708
Recommended Use Laundry single dose liquid pack for clothes washing machines.

Supplier Address

Sun Products Corporate Office
60 Danbury Road
Wilton, CT 06897
USA

Chemical Emergency Phone Number Chemtrec 1-800-424-9300 (24 hour)
Poison Control 1-800-222-1222 (24 hour)

Company Emergency Phone Number Prosar: 1-800-565-5597 (24 hours)

General Questions: General - 1-800-776-6702 (Monday – Friday, 8:30AM to 6:00PM EST)
XCONG

2. HAZARDS IDENTIFICATION

ATTENTION

Emergency Overview

Harmful if swallowed
May cause eye irritation
Keep out of the reach of children
May cause central nervous system depression
MAY BE HARMFUL IF ABSORBED THROUGH SKIN
May be harmful if swallowed

Potential Health Effects

See Section 11 for additional Toxicological information.

Ingestion May be harmful if swallowed. Ingestion may cause gastrointestinal irritation, nausea, vomiting and diarrhea. Do not taste or swallow.

Eyes Irritating to eyes. May cause redness, itching, and pain.

Skin Substance may cause slight skin irritation. Prolonged or repeated contact may dry skin and cause irritation. May be absorbed through the skin in harmful amounts.

Inhalation May be harmful if inhaled. May cause irritation of respiratory tract.

Chronic Effects No known effect based on information supplied.

Aggravated Medical Conditions Central nervous system. Preexisting eye disorders. Skin disorders. Respiratory disorders.

Environmental hazard See Section 12 for additional Ecological Information.

3. COMPOSITION/INFORMATION ON INGREDIENTS

Hazardous ingredients as defined by the OSHA Hazard Communication Standard (29CFR § 1910.1200).

Chemical Name	CAS-No	Weight %
C12-C15 Alcohol Ethoxylate	68131-39-5	30-60
BENZENESULFONIC ACID, C10-16-ALKYL DERIVATIVES, SODIUM SALTS	68081-81-2	10-30
Propylene Glycol	57-55-6	7-13
Ethanolamine	141-43-5	1 - 5 %
DISODIUM DISTYRYLBIPHENYL DISULFONATE	27344-41-8	0.1 - 1%

Under normal consumer household use, this product would not constitute a hazardous product by OSHA definition as OSHA is not applicable to household consumer use.

4. FIRST AID MEASURES

General advice	Consult a physician.
Eye contact	Rinse thoroughly with plenty of water for at least 15 minutes and consult a physician.
Skin contact	Wash skin with soap and water.
Inhalation	Move to fresh air.
Ingestion	Call a physician immediately. Never give anything by mouth to an unconscious person. Do NOT induce vomiting.
Notes to physician	Treat symptomatically.

5. FIRE-FIGHTING MEASURES

Flammable Properties	Not flammable.
Flash point	> 201 °F / > 94 °C
Flammability Limits in Air	
Upper Flammability Limit	No information available
Lower Flammability Limit	No information available
Autoignition temperature	No information available
Hazardous Combustion Products	None.
Explosive properties	No information available
Explosion Limits	
Upper (UEL)	No information available
Lower (LEL)	No information available
Suitable Extinguishing Media	Use extinguishing measures that are appropriate to local circumstances and the surrounding environment.
Unsuitable Extinguishing Media	None.
Protective Equipment and Precautions for Firefighters	As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

NFPA	Health Hazard 2	Flammability 1	Stability 0	Physical and chemical hazards -
HMIS	Health Hazard 2	Flammability 1	Physical Hazard 0	Personal protection -

6. ACCIDENTAL RELEASE MEASURES

Personal precautions	Ensure adequate ventilation.
Environmental precautions	Try to prevent the material from entering drains or water courses.
Methods for Containment	Prevent further leakage or spillage if safe to do so.
Methods for cleaning up	Use personal protective equipment. Dam up. Cover liquid spill with sand, earth or other noncombustible absorbent material. Take up mechanically and collect in suitable container for disposal. Clean contaminated surface thoroughly.
Other information	not applicable.

7. HANDLING AND STORAGE

Advice on safe handling	Handle in accordance with good industrial hygiene and safety practice.
Technical measures/Storage conditions	Keep containers tightly closed in a dry, cool and well-ventilated place.

8. EXPOSURE CONTROLS/PERSONAL PROTECTION

Exposure Guidelines

Chemical Name	ACGIH TLV	OSHA PEL	NIOSH IDLH
Ethanolamine 141-43-5	STEL: 6 ppm TWA: 3 ppm	TWA: 3 ppm TWA: 6 mg/m ³ (vacated) TWA: 3 ppm (vacated) TWA: 8 mg/m ³ (vacated) STEL: 6 ppm (vacated) STEL: 15 mg/m ³	IDLH: 30 ppm TWA: 3 ppm TWA: 8 mg/m ³ STEL: 6 ppm STEL: 15 mg/m ³
Subtilisin 9014-01-1	Ceiling: 0.00006 mg/m ³ as crystalline active enzyme	(vacated) STEL: 0.00006 mg/m ³ 60 min	STEL: 0.00006 mg/m ³ 60 min Subtilisins

NIOSH IDLH: Immediately Dangerous to Life or Health

Other Exposure Guidelines	Vacated limits revoked by the Court of Appeals decision in AFL-CIO v. OSHA, 965 F.2d 962 (11th Cir., 1992).
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Engineering Measures	None under normal use conditions
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Personal Protective Equipment

Eye/Face Protection

Skin and body protection

Respiratory protection

No protective equipment is needed under normal use conditions.
No protective equipment is needed under normal use conditions.
If exposure limits are exceeded or irritation is experienced, NIOSH/MSHA approved respiratory protection should be worn. Positive-pressure supplied air respirators may be required for high airborne contaminant concentrations. Respiratory protection must be provided in accordance with current local regulations.

Hygiene measures	Remove and wash contaminated clothing before re-use. Do not eat, drink or smoke when using this product.
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9. PHYSICAL AND CHEMICAL PROPERTIES

Physical state	liquid	Appearance	LIQUID PACK
Color	blue	Odor	ORIGINAL
Odor Threshold	No information available	pH	7.6 pH.

Specific Gravity	1.02	Density	no data available
Viscosity, kinematic	160 cps	Flash point	> 201 °F / > 94 °C
Method	No information available	Autoignition temperature	No information available
Decomposition temperature	no data available	Boiling point/boiling range	no data available
Melting point	no data available	Freezing Point	no data available
Water solubility	Soluble in water	Evaporation rate	no data available
Vapor pressure	no data available	Vapor density	No data available
VOC Content(%)	16.35		
Flammability Limits in Air		Explosion Limits	
Upper Flammability Limit	no data available	Upper (UEL)	no data available
Lower Flammability Limit	no data available	Lower (LEL)	no data available

10. STABILITY AND REACTIVITY

Stability	Stable under recommended storage conditions.
Incompatible products	Strong oxidizing agents, strong acids, and strong bases.
Conditions to Avoid	Extremes of temperature and direct sunlight. Do not freeze.
Hazardous Decomposition Products	None known based on information supplied.
Hazardous Reactions	None known based on information supplied.
Hazardous Polymerization	Hazardous polymerization does not occur.

11. TOXICOLOGICAL INFORMATION

Acute toxicity

Ingestion	Harmful if swallowed. Ingestion may cause gastrointestinal irritation, nausea, vomiting and diarrhea. Do not taste or swallow.
Eye contact	Irritating to eyes. Avoid contact with eyes.
Skin contact	Causes skin irritation. Prolonged contact may cause redness and irritation. Prolonged or repeated contact may dry skin and cause irritation.
Inhalation	May cause irritation of respiratory tract.
Product Safety Profile	Product is safe for its intended use based on the formulation, testing results, and the long history of safe consumer use.
LD50 Oral:	> 2000 mg/kg (rat)
LD50 Dermal:	> 2000 mg/kg

Chemical Name	LD50 Oral	LD50 Dermal	LC50 Inhalation
C12-C15 Alcohol Ethoxylate	2 g/kg (Rat)	2 g/kg (Rabbit)	
BENZENESULFONIC ACID, C10-16-ALKYL DERIVATIVES, SODIUM SALTS	1000 mg/kg (Rat)	1000 mg/kg (Rabbit)	
Propylene Glycol	20000 mg/kg (Rat)	20800 mg/kg (Rabbit)	
Ethanolamine	1720 mg/kg (Rat)	1 mL/kg (Rabbit) 1025 mg/kg (Rabbit)	
DISODIUM DISTYRYLBIPHENYL DISULFONATE	2000 mg/kg (Rat)	2500 mg/kg (Rabbit) 1 g/kg (Rat)	3.6 mg/L (Rat) 4 h

Chronic toxicity

Chronic toxicity	No known effect based on information supplied. May produce an allergic reaction.
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Carcinogenicity	Contains no ingredients above reportable quantities listed as a carcinogen
Sensitization	None known.
Germ Cell Mutagenicity	None known.
Reproductive toxicity	This product does not contain any known or suspected reproductive hazards
Developmental Toxicity	None known
Target Organ Effects	Central nervous system, Eyes, Respiratory system, Skin
Interactions with Other Chemicals	Use of alcoholic beverages may enhance toxic effects

12. ECOLOGICAL INFORMATION

Ecotoxicity

The environmental impact of this product has not been fully investigated

Chemical Name	Toxicity to algae	Toxicity to fish	Toxicity to microorganisms	Toxicity to daphnia and other aquatic invertebrates
Propylene Glycol	19000: 96 h Pseudokirchneriella subcapitata mg/L EC50	51600: 96 h Oncorhynchus mykiss mg/L LC50 static 41 - 47: 96 h Oncorhynchus mykiss mL/L LC50 static 51400: 96 h Pimephales promelas mg/L LC50 static 710: 96 h Pimephales promelas mg/L LC50		10000: 24 h Daphnia magna mg/L EC50 1000: 48 h Daphnia magna mg/L EC50 Static
Ethanolamine	15: 72 h Desmodesmus subspicatus mg/L EC50	227: 96 h Pimephales promelas mg/L LC50 flow-through 3684: 96 h Brachydanio rerio mg/L LC50 static 300 - 1000: 96 h Lepomis macrochirus mg/L LC50 static 114 - 196: 96 h Oncorhynchus mykiss mg/L LC50 static 200: 96 h Oncorhynchus mykiss mg/L LC50 flow-through	EC50 = 110 mg/L 17 h EC50 = 12200 mg/L 2 h EC50 = 13.7 mg/L 30 min	65: 48 h Daphnia magna mg/L EC50
DISODIUM DISTYRYLBIPHENYL DISULFONATE	10: 72 h Desmodesmus subspicatus mg/L EC50 10.0 - 11.0: 96 h Desmodesmus subspicatus mg/L EC50	76: 96 h Brachydanio rerio mg/L LC50 static		1000: 48 h Daphnia magna mg/L EC50

Persistence and degradability No data is available on the product itself.

Bioaccumulation Not likely to bioaccumulate.

Mobility Not Determined .

Chemical Name	log Pow
Ethanolamine	-1.91

13. DISPOSAL CONSIDERATIONS

Waste Disposal Methods This material, as supplied, is not a hazardous waste according to Federal regulations (40 CFR 261). This material could become a hazardous waste if it is mixed with or otherwise comes in contact with a hazardous waste, if chemical additions are made to this material, or if the material is processed or otherwise altered. Consult 40 CFR 261 to determine whether the altered material is a hazardous waste. Consult the appropriate state, regional, or local regulations for additional requirements

Contaminated packaging Do not re-use empty containers

14. TRANSPORT INFORMATION

DOT Not regulated

TDG Not regulated

MEX Not regulated

IATA Not regulated

IMDG Not regulated

15. REGULATORY INFORMATION

International Inventories

TSCA Complies

DSL Complies

TSCA - United States Toxic Substances Control Act Section 8(b) Inventory
DSL/NDL - Canadian Domestic Substances List/Non-Domestic Substances List

U.S. Federal Regulations

SARA 313

Section 313 of Title III of the Superfund Amendments and Reauthorization Act of 1986 (SARA). This product does not contain any chemicals which are subject to the reporting requirements of the Act and Title 40 of the Code of Federal Regulations, Part 372.

SARA 311/312 Hazard Categories

Acute Health Hazard	Yes
Chronic Health Hazard	no
Fire Hazard	no
Sudden Release of Pressure Hazard	no
Reactive Hazard	no

Clean Water Act

This product does not contain any substances regulated as pollutants pursuant to the Clean Water Act (40 CFR 122.21 and 40 CFR 122.42).

CERCLA

This material, as supplied, does not contain any substances regulated as hazardous substances under the Comprehensive Environmental Response Compensation and Liability Act (CERCLA) (40 CFR 302) or the Superfund Amendments and Reauthorization Act (SARA) (40 CFR 355). There may be specific reporting requirements at the local, regional, or state level pertaining to releases of this material.

U.S. State Regulations

California Proposition 65

This product does not contain any Proposition 65 chemicals.

U.S. State Right-to-Know Regulations

Chemical Name	Massachusetts	New Jersey	Pennsylvania	Illinois	Rhode Island
Propylene Glycol		X	X		
Ethanolamine	X	X	X	X	

Mexico

Mexico - Grade Slight risk, Grade 1

Chemical Name	Carcinogen Status	Exposure Limits
Ethanolamine		Mexico: TWA 3 ppm Mexico: TWA 8 mg/m ³ Mexico: STEL 6 ppm Mexico: STEL 15 mg/m ³

Canada

This product has been classified in accordance with the hazard criteria of the Controlled Products Regulations (CPR) and the MSDS contains all the information required by the CPR.

WHMIS Hazard Class

D2B Toxic materials



16. OTHER INFORMATION

Prepared By Sun Products Corporation
Regulatory Affairs

Revision Date 29-Oct-2013
Revision Note no data available.
Status -5
PROTYP F50

Disclaimer

The (M)SDS is intended to provide a brief summary of our knowledge and guidance regarding the use of this material. The information contained here has been compiled from sources considered by Sun Products Corporation to be dependable and is accurate to the best of the Company's knowledge. It is not meant to be an all-inclusive document on worldwide hazard communication regulations.

This information is offered in good faith. Each user of this material needs to evaluate the conditions of use and design the appropriate protective mechanisms to prevent employee exposures, property damage or release to the environment. Sun Products Corporation assumes no responsibility for injury to the recipient of third persons, or for any damage to any property resulting from misuse of the product.

End of Material Safety Data Sheet